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INFORMS is located in Maryland, USA

Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

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To cite this article:
Minseok Ryu; , Ruiwei Jiang (2025) Nurse Staffing Under Absenteeism: A Distributionally Robust Optimization Approach. Manufacturing & Service Operations Management

Published online in Articles in Advance 09 Jan 2025

. https://doi.org/10.1287/msom.2023.0398

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MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
Articles in Advance, pp. 1–16
ISSN 1523-4614 (print), ISSN 1526-5498 (online)

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Nurse Staffing Under Absenteeism: A Distributionally Robust Optimization Approach

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Received: July 17, 2023
Revised: February 19, 2024; October 15, 2024
Accepted: December 1, 2024
Published Online in Articles in Advance: January 9, 2025

https://doi.org/10.1287/msom.2023.0398

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Abstract. ***Problem definition*:** We study a nurse staffing problem under random nurse demand and absenteeism. Although the demand uncertainty is exogenous, the absenteeism uncertainty is *decision-dependent*, that is, the number of nurses who show up for work partially depends on the nurse staffing level. For quality of care, hospitals develop float pools of hospital units and train nurses to be able to work in multiple units (termed cross-training) in response to potential nurse shortages. ***Methodology/results*:** We study a distributionally robust nurse staffing (DRNS) model that considers both exogenous and *decision-dependent* uncertainties. We derive a separation algorithm to solve this model under a general structure of float pools. In addition, we identify several pool structures that often arise in practice and recast the corresponding DRNS model as a mixed-integer linear program, which facilitates off-the-shelf commercial solvers. ***Managerial implications*:** Through the numerical case studies, based on the data of a collaborating hospital, we found that modeling decision-dependent absenteeism improves the out-of-sample performance of staffing decisions, and such improvement is positively correlated with the value of flexibility arising from fully utilizing float pools.

Funding: R. Jiang was supported in part by the National Science Foundation [Grant ECCS-1845980].
Supplemental Material: The online appendix is available at https://doi.org/10.1287/msom.2023.0398.

Keywords: nurse staffing • decision-dependent uncertainty • distributionally robust optimization • strong valid inequalities • convex hull

1. Introduction

Nurse staffing plays a key role in hospital management. The cost of staffing nurses accounts for over 30% of the overall hospital annual expenditures (see, e.g., Welton 2011). Besides, the nurse staffing level can have significant impact on patient safety, quality of care, and the job satisfaction of nurses (see, e.g., Tevington 2011). In view of that, a number of governing agencies (e.g., California Department of Health 2004 and Victoria Department of Health 2015) have set up minimum nurse-to-patient ratios (NPRs) for various types of hospital units to regulate the staffing decision.

Nurse staffing takes place weeks to months ahead of a specific shift and determines the staffing levels of multiple hospital units based on, for example, the patient census forecast and NPRs. If a unit is found short of nurses because of absence (e.g., family and sick leaves) before the shift, additional workforce such as temporary or off-duty nurses can be recruited. We note that, although nurse staffing does not generate shift schedules and assignments for individual nurses, the output of this study can be used when making those decisions (see Hulshof et al. 2012, Bam et al. 2020).

Nurse staffing is challenging because of the uncertainties of nurse demand and absenteeism. The demand uncertainty has been well documented (Gaynor and Anderson 1995, Boutsioli 2010) and studied in the nurse staffing literature (see, e.g., Davis et al. 2014, Kim and Mehrotra 2015). In contrast, the absenteeism uncertainty has received relatively less attention (see Green et al. 2013, Maass et al. 2017), albeit commonly observed in practice. According to the U.S. Bureau of Labor Statistics (2016), the average absence rate among all nurses in the Veterans Affairs Healthcare System is 6.4%, significantly higher than that among all occupations (2.9%). In response, hospitals develop float pools consisting of multiple hospital units, and a nurse assigned to a pool is cross-trained to be able to work in those units. Hence, a pool nurse can also reinforce when a unit is short of nurses (Inman et al. 2005).

Unlike demand, the nurse absenteeism uncertainty is *decision-dependent*. Specifically, if a unit is staffed with $w \in \mathbb{N}_+$ nurses, then the nurse absenteeism rate depends on w (see Figure 1). Although failing to incorporate such endogeneity may result in understaffing (see Green et al. 2013), unfortunately, modeling endogeneity usually

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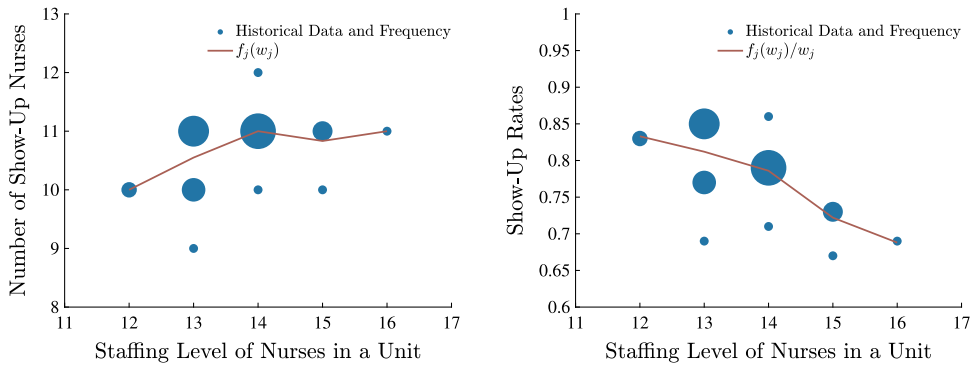
Figure 1. (Color online) Number of Show-Up Nurses and Show-Up Rates, as Functions of Staffing Levels, Based on the Data from a Collaborating Hospital



Note. Dots represent historical data samples, and the size of dots represents frequency.

makes optimization models computationally prohibitive (see Dupačová 2006). Because of this technical difficulty, existing works often resort to exogenous uncertainty for an approximation or employ certain parametric probability distributions, for example, the absence of each nurse follows an independent Bernoulli distribution with the same probability (Green et al. 2013). A basic challenge to parametric models is that our knowledge of the decision-dependent probability distribution is usually *ambiguous*. Under most circumstances, we only have historical data, for example, past nurse staffing levels and absence records, which can be considered as samples taken from the ambiguous distribution. As a result, the solution obtained from a parametric model can yield unpleasant out-of-sample performance if the chosen model is biased.

In this paper, we propose a *nonparametric* model for both exogenous and decision-dependent uncertainties based on distributionally robust optimization (DRO). Our approach considers an ambiguity set of probability distributions based only on the support and moment information of these uncertainties. To the best of our knowledge, this is the first study of the decision-dependent uncertainty in nurse staffing by using a DRO approach.

1.1. Literature Review

A majority of the nurse staffing literature focuses on deterministic models (see Van den Bergh et al. 2013). For example, Kao and Tung (1981) assessed the need for hiring permanent staffs and temporary helpers and Trivedi (1981) analyzed the trade-offs among hiring full-time, part-time, and overtime nurses. More recently, Yang et al. (2007) compared cross-training and flexible work days and demonstrated that cross-training is far more effective for performance improvement than flexible work days. Similarly, Bam et al. (2020) identified cross-training as a promising extension from their deterministic model. Despite the potential benefit of flexibility brought by float pools and cross-training, Maenhout and Vanhoucke (2013) pointed out that the pool design and staffing are often done manually in a qualitative fashion (also see Stewart et al. 1994). In addition, when the nurse demand and/or absenteeism is random, the deterministic models may underestimate the total staffing cost (see, e.g., Kao and Queyranne 1985).

Existing stochastic nurse staffing models often consider the demand uncertainty only. For example, Campbell (2011) studied a two-stage stochastic programming (SP) model that integrates the staffing and scheduling of cross-trained workers under demand uncertainty and demonstrated the value of cross-training. Likewise, using a two-stage SP model under demand uncertainty, Lu and Lu (2016) studied how mandatory overtime laws can negatively affect the service quality of a nursing home. Different from these works, Green et al. (2013) considered random nurse absenteeism. Based on the data from various hospitals, Green et al. (2013) found that the absence rate depends on the staffing level (also see Figure 1) and ignoring such dependency results in understaffing. They successfully derived a closed-form optimal staffing level for a single unit. In addition, Slaugh et al. (2018) derived an optimal staffing level for a single on-call pool. The problem becomes computationally prohibitive when there are multiple units and/or float pools. For example, Easton (2014) studied a multiunit and one-pool setting and pointed out that the proposed SP model does not scale well, even when staffing levels are fixed. Wang and Gupta (2014) considered a multiunit, no-pool setting and found it computationally demanding to search for an optimal staffing strategy. Consequently, both Easton (2014) and Wang and Gupta (2014) resort to easy-to-use heuristics. SP models may also contrast in the modeling of nurse absenteeism. For example, Green et al. (2013) and Easton (2014) assumed that all nurses are absent independently and with a homogeneous rate, whereas Wang and Gupta (2014) suggested that nurses have heterogeneous absence rates. In addition, the nurse absence

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may be correlated because of hidden confounders (e.g., weather and holidays). Consequently, a biased model of the nurse absenteeism may yield suboptimal staffing decisions.

In this paper, we take into account both nurse demand and absenteeism uncertainty in a multiunit and multipool setting. Our DRO model allows heterogeneous absence rates and arbitrary correlations among nurses. Moreover, our model can be solved to global optimum by a separation algorithm and, in several important special cases, by solving a single mixed-integer linear program (MILP).

DRO models have received increasing attention in the recent literature. In particular, DRO has been applied to two-stage stochastic optimization (see, e.g., Bertsimas et al. 2010, 2018; Kong et al. 2013; Hanasusanto and Kuhn 2018), as in this paper. Nevertheless, two-stage DRO is in general computationally prohibitive and proved to be NP-hard by Bertsimas et al. (2010). As compared with this stream of literature, we exploit the nurse staffing problem structure and derive tractable reformulations under practical float pool structures. DRO has also been applied to model decision-dependent uncertainty. Luo and Mehrotra (2020) derived equivalent reformulations of a decision-dependent DRO model under various ambiguity sets. Decision-dependent DRO models have been applied by Noyan et al. (2022) on machine scheduling using an earth mover ambiguity set, by Doan (2022) on retrofitting planning using a Fréchet ambiguity set, and by Basciftci et al. (2021) on facility location using a moment ambiguity set. As compared with the most relevant setting by Basciftci et al. (2021), our staffing decisions impact not only the moments of the random variables but also their support, whose size grows exponentially as the staffing level increases, in contrast to their constant-sized support. We summarize our main contributions as follows:

1. We propose a first decision-dependent DRO approach for nurse staffing, considering both exogenous nurse demand and decision-dependent nurse absenteeism. Our model considers multiple units, multiple float pools, and both long-term and preshift nurse staffing. For general pool structures, we derive a separation algorithm that solves this model to global optimum.

2. For various pool structures arising in practice, we provide a MILP reformulation of the distributionally robust nurse staffing (DRNS) model by deriving an LP representation for a convex maximization problem. In these cases, the computational burden of DRNS is the same as that of the deterministic nurse staffing.

3. We conduct extensive case studies using data from our collaborating hospital. The results demonstrate the value of modeling nurse absenteeism and the computational efficacy of DRNS.

The remainder of the paper is organized as follows. Section 2 describes the DRNS model, Section 3 derives a general solution approach, and Section 4 derives tractable reformulations under special pool structures. We conduct case studies in Section 5, and conclude in Section 6. To ease the exposition, we relegate all proofs to the online appendices.

1.2. Notation

We use $\tilde{}$ to indicate random variables and $\wedge$ to indicate realizations of the random variables. For example, $\tilde{d}$ represents a random variable and $\hat{d}^1, \ldots, \hat{d}^N$ represent N realizations of $\tilde{d}$. For $a, b \in \mathbb{Z}$, we define $[a] := \{1, 2, \ldots, a\}$ and $[a, b]_{\mathbb{Z}} := \{n \in \mathbb{Z} : a \le n \le b\}$. For $x \in \mathbb{R}$, we define $[x]_+ := \max\{x, 0\}$. For set S, we define its indicator function $\mathbb{1}_S$ such that $\mathbb{1}_S(s) := 1$ if $s \in S$ and $\mathbb{1}_S(s) := 0$ if $s \notin S$, and denote its convex hull by $\mathrm{conv}(S)$. We define $\mathbb{B} := \{0, 1\}$.

2. Distributionally Robust Nurse Staffing

For a specific shift, we consider staffing a group of J hospital units, each facing a random demand of nurses denoted by $\tilde{d}_j$ for all $j \in [J]$. To enhance flexibility, we form I nurse float pools. Each pool $i \in [I]$ is associated with a set P_i of units and each nurse assigned to this pool is capable of working in any unit $j \in P_i$. Because of absenteeism, if we staff unit j with w_j nurses (termed unit nurses), then there will be a random number $\tilde{w}_j$ of nurses showing up for work, where $\tilde{w}_j \in [0, w_j]_{\mathbb{Z}}$. Likewise, $\tilde{y}_i$ nurses show up if we staff pool i with y_i nurses during the shift, where $\tilde{y}_i \in [0, y_i]_{\mathbb{Z}}$. After $\tilde{d}_j$, $\tilde{w}_j$, and $\tilde{y}_i$ are realized, the nurses showing up in pool i can be reassigned to any units in P_i to make up the nurse shortage, if any. At that point, any remaining shortage will be covered by hiring temporary nurses in order to meet the NPR requirement. Formally, for given $\tilde{w} := [\tilde{w}_1, \ldots, \tilde{w}_J]^\top$, $\tilde{y} := [\tilde{y}_1, \ldots, \tilde{y}_I]^\top$, and $\tilde{d} := [\tilde{d}_1, \ldots, \tilde{d}_J]^\top$, the cost of hiring temporary nurses can be obtained from an integer program

$$V(\tilde{w}, \tilde{y}, \tilde{d}) := \min_{z \in \mathbb{Z}_+^{I \times |P_i|}, x \in \mathbb{Z}_+^J, e \in \mathbb{Z}_+^J} \quad \sum_{j=1}^{J} c_j^x x_j \tag{1a}$$

$$\text{s.t.} \quad \sum_{i \in [I]: j \in P_i} z_{ij} + x_j - e_j = \tilde{d}_j - \tilde{w}_j, \quad \forall j \in [J], \tag{1b}$$

$$\sum_{j \in P_i} z_{ij} \le \tilde{y}_i, \quad \forall i \in [I], \tag{1c}$$

where variables z_{ij} represent the number of nurses reassigned from pool i to unit j, variables x_j represent the number of temporary nurses[1] hired, variables e_j represent the excessive number of nurses, and parameters c_j^x represent the unit cost of hiring temporary nurses. In the above formulation, constraints (1b) describe three ways of satisfying the demand in unit j, (i) assigning unit nurses, (ii) reassigning pool nurses, and (iii) hiring temporary nurses, and constraints (1c) ensure that the

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number of nurses reassigned from pool i does not exceed the number of nurses showing up in that pool.

As mentioned in Section 1, the probability distribution $\mathbb{P}_{\tilde{w},\tilde{y},\tilde{d}}$ of $(\tilde{w},\tilde{y},\tilde{d})$ is ambiguous and depends on the staffing levels (w, y) of units and pools. When historical data of $(\tilde{w},\tilde{y},\tilde{d})$ are available, a natural and interpretable way of modeling such dependence is through its support and moments (e.g., mean values), as a function of (w, y). This motivates us to consider the following decision-dependent ambiguity set $\mathcal{D}(w,y)$:

$$\mathcal{D}(w,y)=\{\mathbb{P}\in\mathcal{P}(\Xi):\mathbb{E}_{\mathbb{P}}[\tilde{d}_j^q]=\mu_{jq},\ \forall j\in[J],\ \forall q\in[Q], \quad (2a)$$

$$\mathbb{E}_{\mathbb{P}}[\tilde{w}_j]=f_j(w_j),\ \forall j\in[J],$$
$$\mathbb{E}_{\mathbb{P}}[\tilde{y}_i]=g_i(y_i),\ \forall i\in[I]\}, \quad (2b)$$

where Ξ represents the support of $(\tilde{w},\tilde{y},\tilde{d})$ and $\mathcal{P}(\Xi)$ represents the set of probability distributions supported on Ξ. We consider a box support $\Xi := \Xi_{\tilde{w}} \times \Xi_{\tilde{y}} \times \Xi_{\tilde{d}}$, where $\Xi_{\tilde{w}} = \Pi_{j=1}^{J}[0,w_j]_{\mathbb{Z}}$, $\Xi_{\tilde{y}} = \Pi_{i=1}^{I}[0,y_i]_{\mathbb{Z}}$, $\Xi_{\tilde{d}} = \Pi_{j=1}^{J}[d_j^{\mathrm{L}}, d_j^{\mathrm{U}}]_{\mathbb{Z}}$, and d_j^{L} and d_j^{U} represent lower and upper bounds of the nurse demand in unit j. In addition, μ_{jq} represents the q-th moment of $\tilde{d}_j$ for all $q\in[Q]$. Furthermore, $f_j:\mathbb{Z}_+\to\mathbb{R}_+$ and $g_i:\mathbb{Z}_+\to\mathbb{R}_+$ are two piecewise linear functions that model the dependence of the mean values $(\mathbb{E}_{\mathbb{P}}[\tilde{w}],\mathbb{E}_{\mathbb{P}}[\tilde{y}])$ on the staffing decisions (w, y).

The ambiguity set $\mathcal{D}(w,y)$ can be calibrated conveniently. Suppose that $\mathbb{P}_{\tilde{w},\tilde{y},\tilde{d}}$ is observed through nurse demand data $\{\hat{d}_j^1,\ldots,\hat{d}_j^N\}_{j=1}^J$ and attendance records $\{(w_j^1,\hat{w}_j^1),\ldots,(w_j^N,\hat{w}_j^N)\}_{j=1}^J$ and $\{(y_i^1,\hat{y}_j^1),\ldots,(y_i^N,\hat{y}_i^N)\}_{i=1}^I$ during the past N days, where, in each pair $(w_j^n,\hat{w}_j^n)$, w_j^n represents the staffing level of unit j in day n and $\hat{w}_j^n$ represents the corresponding number of nurses who actually showed up. Then, μ_{jq} can be obtained from empirical estimates (e.g., $\mu_{j1}=(1/N)\sum_{n=1}^N\hat{d}_j^n$, $\mu_{j2}=(1/N)\sum_{n=1}^N(\hat{d}_j^n)^2$, and so on). Furthermore, f_j and g_i can be obtained by performing segmented linear regression on the attendance data, using the staffing levels $\{w_j^1,\ldots,w_j^N\}$ and $\{y_i^1,\ldots,y_i^N\}$ as breakpoints, respectively (see the red solid curves in Figure 1 (left) for an example).

We seek nurse staffing levels that minimize the expected total cost with regard to the worst-case probability distribution in $\mathcal{D}(w,y)$, that is, we consider the following two-stage DRO model:

$$(\textbf{DRNS}):\ \min_{w,y}\ \sum_{j=1}^{J}c_j^{\mathrm{w}}w_j+\sum_{i=1}^{I}c_i^{\mathrm{y}}y_i+\max_{\mathbb{P}\in\mathcal{D}(w,y)}\mathbb{E}_{\mathbb{P}}[V(\tilde{w},\tilde{y},\tilde{d})] \quad (3a)$$

$$\text{s.t. } w\in[w^{\mathrm{L}},w^{\mathrm{U}}]_{\mathbb{Z}},\ y\in[y^{\mathrm{L}},y^{\mathrm{U}}]_{\mathbb{Z}}, \quad (3b)$$

$$w,y\in R\cap\mathbb{Z}_+^{I+J}, \quad (3c)$$

where parameters c^{w} and c^{y} represent the unit cost of hiring unit and pool nurses, respectively; constraints (3b) designate lower and upper bounds on staffing levels, and set R represents all remaining restrictions on (w, y), which we assume can be represented via mixed-integer linear inequalities. We remark that (DRNS) decides the staffing levels during a specific shift and does not decide the schedule for individual nurses across multiple shifts. Hence, multistage models do not apply to our problem. Nevertheless, (DRNS) is computationally challenging because (i) $\mathcal{D}(w,y)$ involves exponentially many decision-dependent probability distributions and (ii) it is a two-stage DRO model with integer recourse variables. In the next two sections, we shall derive equivalent reformulations of (DRNS) that facilitate a separation algorithm, and identify practical pool structures that admit more tractable solution approaches.

The ambiguity set $\mathcal{D}(w,y)$ can be extended by incorporating other moment information of $\mathbb{P}_{\tilde{w},\tilde{y},\tilde{d}}$, for example, the correlations of the random demands (see Delage and Ye 2010). But unfortunately, in our case this produces an intractable formulation (see theorem 3.1 in Bertsimas et al. 2010). Nevertheless, in Section 5.5 we shall incorporate higher marginal moments (e.g., variance, skewness, and kurtosis) of $\tilde{w}$ and $\tilde{y}$ into $\mathcal{D}(w,y)$ and derive a tractable and conservative approximation for the ensuing model.

3. Solution Approach: General Pool Structure

In this section, we consider general pool structures, recast (DRNS) as a min-max formulation, and derive a separation algorithm that solves this model to global optimality.

We start by noticing that the integrality restrictions in the second-stage formulation of (DRNS) can be relaxed without loss of generality.

Lemma 1. *For any given $(\tilde{w},\tilde{y},\tilde{d})\in\Xi$, the value of $V(\tilde{w},\tilde{y},\tilde{d})$ remains unchanged if the integrality restriction is relaxed.*

By Lemma 1 (see Online Appendix A for a proof), we rewrite $V(\tilde{w},\tilde{y},\tilde{d})$ as the following dual formulation:

$$V(\tilde{w},\tilde{y},\tilde{d})=\max_{\alpha,\beta}\ \sum_{j=1}^{J}(\tilde{d}_j-\tilde{w}_j)\alpha_j+\sum_{i=1}^{I}\tilde{y}_i\beta_i \quad (4a)$$

$$\text{s.t. } \beta_i+\alpha_j\le 0,\quad \forall i\in[I],\ \forall j\in P_i, \quad (4b)$$

$$0\le\alpha_j\le c_j^{\mathrm{x}},\quad \forall j\in[J], \quad (4c)$$

where dual variables α_j and β_i are associated with constraints (1b) and (1c), respectively, and constraints (4b) and (4c) are associated with primal variables z_{ij} and (x_j, e_j), respectively. We let Λ denote the dual feasible region, that is, $\Lambda := \{(\alpha,\beta):(4b)\text{–}(4c)\}$. Strong duality between formulations (1a)–(1c) and (4a)–(4c) holds because (1a)–(1c) has a finite optimal value.

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We are now ready to recast (DRNS) as a min-max formulation. To this end, we consider $\mathbb{P}$ as a decision variable and take the dual of the worst-case expectation in (3a). For strong duality, we make the following technical assumption on the ambiguity set $\mathcal{D}(w,y)$.

Assumption 1. *For any staffing levels (w, y) feasible to (DRNS), $\mathcal{D}(w,y)$ is nonempty.*

Assumption 1 is mild. For example, it holds whenever the moments of demands $\{\mu_{jq} : j \in [J], q \in [Q]\}$ are obtained from empirical estimates and the decision-dependent moments $\{g_i(y_i), f_j(w_j) : i \in [I], j \in [J]\}$ lie in the convex hull of their support, that is, $f_j(w_j) \in [0, w_j]$ and $g_i(y_i) \in [0, y_i]$. In Online Appendix B, we present an approach to verify Assumption 1 by solving linear programs. The following proposition presents a min-max reformulation for DRNS (see Online Appendix C for a proof).

Proposition 1. *Under Assumption 1, the (DRNS) model yields the same optimal value and the same set of optimal solutions as the following min-max optimization problem:*

$$\min_{\substack{w,y\\ \gamma,\lambda,\rho}} \max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta) + \sum_{i=1}^{I}(c_i^y y_i + g_i(y_i)\lambda_i) + \sum_{j=1}^{J}\left[c_j^w w_j + \sum_{q=1}^{Q}\mu_{jq}\rho_{jq} + f_j(w_j)\gamma_j\right] \tag{5a}$$

$$\text{s.t. } (3b) - (3c), \tag{5b}$$

where

$$F(\alpha,\beta) := \sum_{j=1}^{J}[(-\alpha_j - \gamma_j)w_j]_+ + \sum_{i=1}^{I}[(\beta_i - \lambda_i)y_i]_+ + \sum_{j=1}^{J}\max_{\tilde{d}_j\in[d_j^L, d_j^U]_{\mathbb{Z}}}\left\{\alpha_j\tilde{d}_j - \sum_{q=1}^{Q}\rho_{jq}\tilde{d}_j^q\right\}. \tag{5c}$$

In the min-max reformulation (5a)–(5b), the additional variables γ, λ, ρ are generated in the process of taking dual. In addition, function $F(\alpha,\beta)$ is jointly convex in (α,β) because, as presented in (5c), $F(\alpha,\beta)$ is the pointwise maximum of functions affine in (α,β). This min-max reformulation is not directly computable because (i) evaluating the objective function (5a) requires us to solve a *convex maximization* problem $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$, which is in general NP-hard; and (ii) the formulation includes nonlinear and nonconvex terms $g_i(y_i)\lambda_i$ and $f_j(w_j)\gamma_j$. In what follows, we shall address these two challenges before presenting a separation algorithm for solving (DRNS).

First, we analyze the convex maximization problem and derive optimality conditions in Lemma 2 (see Online Appendix D for a proof). We assume $0 \le c_1^x \le \cdots \le c_J^x$ without loss of generality.

Lemma 2. *There exists an optimal solution $(\overline{\alpha},\overline{\beta})$ to problem $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ such that (a) $\overline{\alpha}_j \in \{0, c_1^x, \ldots, c_j^x\}$ for all $j \in [J]$ and (b) $\overline{\beta}_i = -\max\{\overline{\alpha}_j : j \in P_i\}$ for all $i \in [I]$.*

Remark 1 (Interpretation on $\overline{\alpha}$ and $\overline{\beta}$). $\overline{\alpha}$ are the shadow prices associated with constraints (1b), that is, $\overline{\alpha}_j$ evaluates the increase in staffing cost when the nurse shortage $(\tilde{d}_j - \tilde{w}_j)$ of unit j increases by one. When this happens, a new optimal solution to formulation (1) reassigns a pool nurse from a unit $k \in \{1, \ldots, j\}$ to unit j because of the ascending staffing costs $c_1^x \le \cdots \le c_j^x$. As a result, it is equivalent to transferring the additional shortage from unit j to unit k and accordingly the staffing cost increases by $\overline{\alpha}_j \in \{0, c_1^x, \ldots, c_j^x\}$. In addition, $\overline{\beta}_i$ evaluates the decrease in staffing cost when an additional nurse shows up in pool i. When that happens, a new optimal solution to (1) assigns this nurse to a unit in which the shortage is most costly. As a result, $-\overline{\beta}_i = \max\{\overline{\alpha}_j : j \in P_i\}$.

Lemma 2 enables us to avoid enumerating the infinite number of elements in Λ and focus only on a finite set of $(\overline{\alpha},\overline{\beta})$ values. In addition, we introduce binary variables to encode the special structure identified in the optimality conditions. Specifically, for all $j \in [J]$ and $k \in [j]$, we define binary variable t_{jk} such that $t_{jk} := 1$ if $\overline{\alpha}_j = c_k^x$, and $t_{jk} := 0$ otherwise. For all $i \in [I]$, we denote by $J(i)$ the largest index in P_i, that is, $J(i) = \max\{j : j \in P_i\}$. For all $i \in [I]$ and $k \in [J(i)]$, we define binary variable $s_{ik} := 1$ if $\overline{\beta}_i = -c_k^x$ and $s_{ik} := 0$ otherwise. In other words, $s_{ik} = 1$ if k is the largest index in $[J(i)]$ with $t_{jk} = 1$ for some $j \in P_i$, in line with Condition (b) in Lemma 2. Variables (t, s) need to satisfy the following constraints to make the encoding well defined:

$$\sum_{k=1}^{j} t_{jk} \le 1, \ \forall j \in [J], \tag{6a}$$

$$\sum_{k=1}^{J(i)} s_{ik} \le 1, \ \forall i \in [I], \tag{6b}$$

$$s_{ik} \le \sum_{j\in P_i: j\ge k} t_{jk}, \ \forall i \in [I], \ \forall k \in [J(i)], \tag{6c}$$

$$s_{ik} + t_{j\ell} \le 1, \ \forall i \in [I], \ \forall k \in [J(i)], \ \forall j \in P_i, \ \forall \ell \in [k+1, j]_{\mathbb{Z}}, \tag{6d}$$

$$t_{jk} \le \sum_{\ell=1}^{J(i)} s_{i\ell}, \ \forall i \in [I], \ \forall j \in P_i, \ \forall k \in [j], \tag{6e}$$

$$s_{ik} \in \mathbb{B}, \ \forall i \in [I], \ \forall k \in [J(i)], \ t_{jk} \in \mathbb{B}, \ \forall j \in [J], \ \forall k \in [j], \tag{6f}$$

where constraints (6a) describe that, for all $j \in [J]$, $t_{jk} = 1$ holds for at most one $k \in [j]$; constraints (6b) describe that $s_{ik} = 1$ holds for at most one $k \in [J(i)]$; constraints (6c) designate that if $s_{ik} = 1$, then there is a $j \in P_i$ with $t_{jk} = 1$; constraints (6d) describe that if $s_{ik} = 1$, then there cannot be any $j \in P_i$ and $\ell \ge k+1$ with $t_{j\ell} = 1$; and

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constraints (6e) ensure that all $t_{jk} = 0$ if $s_{i\ell} = 0$ for all $\ell \in [J(i)]$. This recasts $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ as an integer linear program presented in Theorem 1 (see Online Appendix E for a proof). For ease of exposition, we introduce auxiliary variables $r_j := 1 - \sum_{k=1}^{j} t_{jk}$ and $p_i := 1 - \sum_{k=1}^{J(i)} s_{ik}$, whose values depend entirely on variables t and s.

Theorem 1. *The problem* $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ *yields the same optimal value as the following integer linear program:*

$$\max_{t,s,r,p} \sum_{j=1}^{J}\left(c_j^r r_j + \sum_{k=1}^{j} c_{jk}^t t_{jk}\right) + \sum_{i=1}^{I}\left(c_i^p p_i + \sum_{k=1}^{J(i)} c_{ik}^s s_{ik}\right) \tag{7a}$$

$$\text{s.t. } (t,s,r,p)\in\mathcal{H} := \Big\{(6a)-(6f),\ r_j + \sum_{k=1}^{j} t_{jk} = 1,\ \forall j\in[J],\ \ p_i + \sum_{k=1}^{J(i)} s_{ik} = 1,\ \forall i\in[I]\Big\}, \tag{7b}$$

where $c_j^r := [-\gamma_j w_j]_+ + \max_{\tilde{d}_j\in[d_j^L,d_j^U]_{\mathbb{Z}}}\{-\sum_{q=1}^{Q}\rho_{jq}\tilde{d}_j^q\}$, $c_{jk}^t := [(-c_k^x-\gamma_j)w_j]_+ + \max_{\tilde{d}_j\in[d_j^L,d_j^U]_{\mathbb{Z}}}\{c_k^x\tilde{d}_j - \sum_{q=1}^{Q}\rho_{jq}\tilde{d}_j^q\}$, $c_i^p := [-\lambda_i y_i]_+$, *and* $c_{ik}^s := [(-c_k^x-\lambda_i)y_i]_+$.

Second, we linearize the terms $f_j(w_j)\gamma_j$ and $g_i(y_i)\lambda_i$. For all $j\in[J]$, although $f_j(w_j)$ can be nonlinear and nonconvex, thanks to the integrality of w_j, we can rewrite $f_j(w_j)$ as an affine function based on a binary expansion of w_j. Specifically, we introduce binary variables $\{u_{j\ell} : \ell\in[w_j^U - w_j^L]\}$ such that $w_j = w_j^L + \sum_{\ell=1}^{w_j^U - w_j^L} u_{j\ell}$, where we interpret $u_{j\ell}$ as whether we assign at least $w_j^L + \ell$ nurses to unit j. That is, $u_{j\ell} = 1$ if $w_j \ge w_j^L + \ell$ and $u_{j\ell} = 0$ otherwise. Then, defining $\Delta_{j\ell} := f_j(w_j^L + \ell) - f_j(w_j^L + \ell - 1)$ for all $\ell\in[w_j^U - w_j^L]$, we have

$$\begin{aligned} f_j(w_j) &= f_j(w_j^L) + \sum_{\ell=1}^{w_j - w_j^L}[f_j(w_j^L+\ell) - f_j(w_j^L+\ell-1)] \\ &= f_j(w_j^L) + \sum_{\ell=1}^{w_j^U - w_j^L}[f_j(w_j^L+\ell) - f_j(w_j^L+\ell-1)]\mathbb{1}_{[w_j^L+\ell, w_j^U]}(w_j) \\ &= f_j(w_j^L) + \sum_{\ell=1}^{w_j^U - w_j^L}\Delta_{j\ell}u_{j\ell}. \end{aligned}$$

It follows that $f_j(w_j)\gamma_j = f_j(w_j^L)\gamma_j + \sum_{\ell=1}^{w_j^U - w_j^L}\Delta_{j\ell}u_{j\ell}\gamma_j$. We can linearize the bilinear terms $u_{j\ell}\gamma_j$ by defining continuous variables $\varphi_{j\ell} := u_{j\ell}\gamma_j$ and incorporating the following standard McCormick inequalities (see McCormick 1976):

$$\gamma_j - M(1-u_{j\ell}) \le \varphi_{j\ell} \le M u_{j\ell},\ \ \forall j\in[J],\ \forall \ell\in[w_j^U - w_j^L], \tag{8a}$$

$$-M u_{j\ell} \le \varphi_{j\ell} \le \gamma_j + M(1-u_{j\ell}),\ \ \forall j\in[J],\ \forall \ell\in[w_j^U - w_j^L], \tag{8b}$$

where M represents a sufficiently large positive constant. Likewise, we rewrite $g_i(y_i)\lambda_i$ as $g_i(y_i^L)\lambda_i + \sum_{\ell=1}^{y_i^U - y_i^L}\delta_{i\ell}v_{i\ell}\lambda_i$ by using constants $\delta_{i\ell} := g_i(y_i^L+\ell) - g_i(y_i^L+\ell-1)$ for all $\ell\in[y_i^U - y_i^L]$ and binary variables $\{v_{i\ell} : \ell\in[y_i^U - y_i^L]\}$, where $v_{i\ell} = 1$ if $y_i \ge y_i^L + \ell$ and $v_{i\ell} = 0$ otherwise. We linearize the bilinear terms $v_{i\ell}\lambda_i$ by continuous variables $\nu_{i\ell} := v_{i\ell}\lambda_i$ and the McCormick inequalities

$$\lambda_i - M(1-v_{i\ell}) \le \nu_{i\ell} \le M v_{i\ell},\ \forall i\in[I],\ \forall \ell\in[y_i^U - y_i^L], \tag{8c}$$

$$-M v_{i\ell} \le \nu_{i\ell} \le \lambda_i + M(1-v_{i\ell}),\ \forall i\in[I],\ \forall \ell\in[y_i^U - y_i^L]. \tag{8d}$$

In computation, a large coefficient M slows down the solution of (DRNS). Theoretically, for the correctness of the linearization (8a)–(8d), M needs to be larger than $|\gamma_j|$ and $|\lambda_i|$ for all $j\in[J]$ and $i\in[I]$, respectively. The following proposition (see Online Appendix F for a proof) derives uniform lower and upper bounds of γ_j and λ_i, leading to a small value of M.

Proposition 2. *There exists an optimal solution* $(\gamma^*,\lambda^*,\rho^*)$ *to formulation* (5a)–(5b) *such that* $\gamma_j^* \in [-c_j^x, 0]$ *for all* $j\in[J]$ *and* $\lambda_i^* \in [-c_{J(i)}^x, 0]$ *for all* $i\in[I]$.

Proposition 2 indicates that (i) we can set $M := c_j^x$ in (8a)–(8d) without loss of optimality and (ii) as all γ_j and λ_i are nonpositive at optimality, we can replace McCormick inequalities (8a) and (8c) as $\gamma_j \le \varphi_{j\ell} \le 0$ and $\lambda_i \le \nu_{i\ell} \le 0$, respectively, both of which are now *big-M-free*. In addition, we incorporate the following constraints to break the symmetry among binary variables:

$$u_{j\ell} \ge u_{j(\ell+1)},\ \ \forall j\in[J],\ \forall \ell\in[w_j^U - w_j^L - 1], \tag{8e}$$

$$v_{i\ell} \ge v_{i(\ell+1)},\ \ \forall i\in[I],\ \forall \ell\in[y_i^U - y_i^L - 1]. \tag{8f}$$

The above analysis recasts (DRNS) into a mixed-integer program, which is summarized in the following theorem without proof.

Theorem 2. *Under Assumption 1, the (DRNS) model yields the same optimal value as the following mixed-integer program:*

$$\min_{\substack{u,v,\varphi,\nu\\ \gamma,\lambda,\rho,\theta}} \theta + \sum_{i=1}^{I}\Big(c_i^y y_i^L + g_i(y_i^L)\lambda_i + \sum_{\ell=1}^{y_i^U - y_i^L}(c_i^y v_{i\ell} + \delta_{i\ell}\nu_{i\ell})\Big) + \sum_{j=1}^{J}\Big[\sum_{q=1}^{Q}\mu_{jq}\rho_{jq} + c_j^w w_j^L + f_j(w_j^L)\gamma_j + \sum_{\ell=1}^{w_j^U - w_j^L}(c_j^w u_{j\ell} + \Delta_{j\ell}\varphi_{j\ell})\Big] \tag{9a}$$

$$\text{s.t. } (8a)-(8f), \tag{9b}$$

$$\theta \ge \sum_{j=1}^{J}\left(c_j^r r_j + \sum_{k=1}^{j} c_{jk}^t t_{jk}\right) + \sum_{i=1}^{I}\left(c_i^p p_i + \sum_{k=1}^{J(i)} c_{ik}^s s_{ik}\right),\ \ \forall (t,s,r,p)\in\mathcal{H}, \tag{9c}$$

$$u_{j\ell}\in\mathbb{B},\ \forall j\in[J],\ \forall \ell\in[w_j^U - w_j^L],\ v_{i\ell}\in\mathbb{B},\ \forall i\in[I],\ \forall \ell\in[y_i^U - y_i^L], \tag{9d}$$

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where set $\mathcal{H}$ is defined in (7b) and coefficients c_j^r, c_{jk}^t, c_i^p, and c_{ik}^s are represented through

$$c_j^r = \left[-\gamma_j w_j^L - \sum_{\ell=1}^{w_j^U - w_j^L} \varphi_{j\ell}\right]_+ + \max_{\tilde{d}_j \in [d_j^L, d_j^U]_{\mathbb{Z}}} \left\{-\sum_{q=1}^{Q} \rho_{jq} \tilde{d}_j^q\right\}, \tag{9e}$$

$$c_{jk}^t = \left[(-c_k^x - \gamma_j) w_j^L - \sum_{\ell=1}^{w_j^U - w_j^L} (c_k^x u_{j\ell} + \varphi_{j\ell})\right]_+ + \max_{\tilde{d}_j \in [d_j^L, d_j^U]_{\mathbb{Z}}} \left\{c_k^x \tilde{d}_j - \sum_{q=1}^{Q} \rho_{jq} \tilde{d}_j^q\right\}, \tag{9f}$$

$$c_i^p = \left[-\lambda_i y_i^L - \sum_{\ell=1}^{y_i^U - y_i^L} \nu_{i\ell}\right]_+, \tag{9g}$$

$$c_{ik}^s = \left[(-c_k^x - \lambda_i) y_i^L - \sum_{\ell=1}^{y_i^U - y_i^L} (c_k^x v_{i\ell} + \nu_{i\ell})\right]_+. \tag{9h}$$

Algorithm 1 (A Separation Algorithm for Solving the (DRNS) Model (3))

1: Initialization: Set the set of cuts $\mathcal{H}_{\text{sep}} \leftarrow \emptyset$.
2: Solve the master problem

$$(\mathbf{MP}): \min_{\substack{u,v,\varphi,\nu \\ \gamma,\lambda,\rho}} \theta + \sum_{i=1}^{I} \left(c_i^y y_i^L + g_i(y_i^L)\lambda_i + \sum_{\ell=1}^{y_i^U - y_i^L} (c_i^y v_{i\ell} + \delta_{i\ell} \nu_{i\ell})\right) + \sum_{j=1}^{J} \left[\sum_{q=1}^{Q} \mu_{jq} \rho_{jq} + c_j^w w_j^L + f_j(w_j^L)\gamma_j + \sum_{\ell=1}^{w_j^U - w_j^L} (c_j^w u_{j\ell} + \Delta_{j\ell} \varphi_{j\ell})\right]$$

$$\text{s.t. (9b),(9d)},$$

$$\theta \geq \sum_{j=1}^{J} \left(c_j^r r_j + \sum_{k=1}^{j} c_{jk}^t t_{jk}\right) + \sum_{i=1}^{I} \left(c_i^p p_i + \sum_{k=1}^{J(i)} c_{ik}^s s_{ik}\right), \quad \forall (t,s,r,p) \in \mathcal{H}_{\text{sep}},$$

3: Compute $(c_j^{r*}, c_{jk}^{t*}, c_i^{p*}, c_{ik}^{s*})$ by (9e)–(9h) and $(u^*, v^*, \varphi^*, \nu^*, \gamma^*, \lambda^*, \rho^*)$ by (**MP**).
4: Solve (7) using $(c_j^{r*}, c_{jk}^{t*}, c_i^{p*}, c_{ik}^{s*})$ and record an optimal solution (t^*, s^*, r^*, p^*).
5: **if** $\theta^* \geq \sum_{j=1}^{J} (c_j^{r*} r_j^* + \sum_{k=1}^{j} c_{jk}^{t*} t_{jk}^*) + \sum_{i=1}^{I} (c_i^{p*} p_i^* + \sum_{k=1}^{J(i)} c_{ik}^{s*} s_{ik}^*)$ **then**
6: Stop and return (u^*, v^*) as an optimal solution to (DRNS).
7: **else**
8: Add a cut (9c) into (**MP**) by setting $\mathcal{H}_{\text{sep}} \leftarrow \mathcal{H}_{\text{sep}} \cup \{(t^*, s^*, r^*, p^*)\}$. Go to step 2.
9: **end if**

The reformulation (9a)–(9d) facilitates the separation algorithm (see, e.g., Nemhauser and Wolsey 1999), also known as delayed constraint generation. We notice that (9c) involves $(J+1)!$ many constraints, making it computationally prohibitive to solve (9a)–(9d) in one shot. Instead, the separation algorithm incorporates constraints (9c) on the fly. Specifically, this algorithm first solves a relaxation of the reformulation by overlooking constraints (9c). Then, we check if the optimal solution thus obtained violates any of (9c). If yes, then we add one violated constraint back into the relaxation and resolve. We call this added constraint a "cut" and note that each cut describes a convex feasible region. This procedure is repeated until an optimal solution is found to satisfy all of constraints (9c). We summarize this procedure in Algorithm 1.

Theorem 3. *Algorithm 1 finds a globally optimal solution to the (DRNS) model (3) in a finite number of iterations.*

To strengthen the integer program (7), we derive valid inequalities for its feasible region in Lemma 3 (see Online Appendix H for a proof).

Lemma 3. *Any $(t,s,r,p) \in \mathcal{H}$ satisfies the following inequalities:*

$$\sum_{\ell=k}^{j} t_{j\ell} \leq \sum_{\ell=k}^{J(i)} s_{i\ell}, \qquad \forall i \in [I],\ \forall j \in P_i,\ \forall k \in [j], \tag{10}$$

where $J(i)$ is the largest index in P_i. In addition, inequalities (10) imply constraints (6e).

Intuitively, inequalities (10) designate that if $\overline{\alpha}_j \geq c_k^x$, then $\overline{\beta}_i \leq -c_k^x$, echoing the optimality conditions on $\overline{\alpha}_j$ and $\overline{\beta}_i$ in Lemma 2. In the following section, we show that, under special but practical pool structures, inequalities (10) are strong enough to describe the convex hull of $\mathcal{H}$. That is, with inequalities (10), we will be able to solve integer program (7) as a linear program and obtain a binary-valued optimal solution. In Section 5.2, we demonstrate that (10) significantly accelerates the solution of (DRNS) under both general and special pool structures.

4. Tractable Cases: Practical Pool Structures

We consider the following three special nurse pool structures.

Structure 1 (One Pool). In this structure, $I = 1$, that is, there is a nurse pool shared among all units (see Figure 2(a) for an example).

Structure D (Disjoint Pools). All nurse pools are disjoint, that is, for all $i_1, i_2 \in [I]$ and $i_1 \neq i_2$, it holds that $P_{i_1} \cap P_{i_2} = \emptyset$ (see Figure 2(b) for an example).

Structure C (Chained Pools). The nurse pools form a long chain, that is, there are $I = J$ pools and the bipartite graph consisting of the units, pools, and pool-unit assignment forms one undirected cycle. Figure 2(c)

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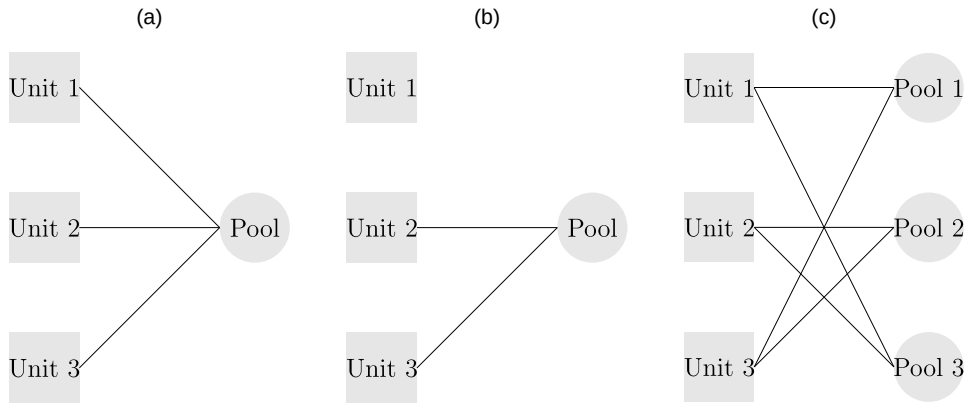
Figure 2. Examples of Practical Pool Structures



Notes. (a) Structure 1. (b) Structure D. (c) Structure C.

displays an example with cycle P_1–Unit 1–P_3–Unit 2–P_2–Unit 3–P_1.

Structure 1 can be utilized when all units have similar functionalities and so they can all share one nurse pool. Accordingly, nurses assigned to this pool need to be cross-trained for all units so that they are able to undertake the tasks in them. Structure D is less demanding than one pool, as each pool covers only a subset of units with similar functionality. Accordingly, the amount of cross-training under this structure significantly decreases. Structure C has been applied in production systems to increase flexibility (see, e.g., Jordan and Graves 1995; Chen et al. 2015, 2019; Wang and Zhang 2015). Under this structure, every unit is covered by two nurse pools. Accordingly, a pool nurse needs to be cross-trained for only two units. All three structures have been considered and compared in a nurse staffing context (see, e.g., Inman et al. 2005). Under these practical pool structures, we derive tractable reformulations of the (DRNS) model. Our derivation leads to MILP reformulations that facilitate powerful off-the-shelf software (e.g., GUROBI).

4.1. One Pool

Under Structure 1, that is, $I = 1$, $P_1 = [J]$, and $J(1) = J$, we rewrite the feasible region $\mathcal{H}$ of integer program (7) as follows:

$$\mathcal{H}_1 = \Bigg\{(t,s,r,p) : r_j + \sum_{k=1}^{j} t_{jk} = 1,\ \forall j \in [J],\ p + \sum_{k=1}^{J} s_k = 1, \tag{11a}$$

$$\sum_{k=1}^{j} t_{jk} \le 1,\ \forall j \in [J], \tag{11b}$$

$$\sum_{k=1}^{J} s_k \le 1, \tag{11c}$$

$$s_k \le \sum_{j=k}^{J} t_{jk},\ \forall k \in [J], \tag{11d}$$

$$s_k + t_{j\ell} \le 1,\ \forall k \in [J],\ \forall j \in [J],\ \forall \ell \in [k+1, j]_{\mathbb{Z}}, \tag{11e}$$

$$t_{jk} \le \sum_{\ell=1}^{J} s_\ell,\ \forall j \in [J],\ \forall k \in [j], \tag{11f}$$

$$s_k \in \mathbb{B},\ \forall k \in [J],\ t_{jk} \in \mathbb{B},\ \forall j \in [J],\ \forall k \in [j] \Bigg\}. \tag{11g}$$

We claim that valid inequalities (10), in conjunction with constraints (11a)–(11d), are sufficient to describe the convex hull of $\mathcal{H}_1$ as described in Proposition 3 (see Online Appendix I for a proof).

Proposition 3. *It holds that* $\bar{\mathcal{H}}_1 = \mathrm{conv}(\mathcal{H}_1)$, *where*

$$\bar{\mathcal{H}}_1 := \Bigg\{(t,s,r,p) : (11a) - (11d),\ \sum_{\ell=k}^{j} t_{j\ell} \le \sum_{\ell=k}^{J} s_\ell,\ \forall j \in [J],\ \forall k \in [j], \tag{12a}$$

$$s_k \in \mathbb{R}_+,\ \forall k \in [J],\ t_{jk} \in \mathbb{R}_+,\ \forall j \in [J],\ \forall k \in [j] \Bigg\}. \tag{12b}$$

Better still, this yields a closed-form solution to the convex maximization problem $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$.

Theorem 4. *Under Structure 1, the optimal value of* $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ *equals*

$$\max\Bigg\{c_1^p + \sum_{j=1}^{J} c_j^r,\ \max_{\substack{j^*\in[J],\\ k^*\in[j^*]}} \Bigg\{\sum_{j=1}^{J} c_j^r + c_{1k^*}^s + (c_{j^*k^*}^t - c_{j^*}^r) + \sum_{j\in[J]\setminus\{j^*\}} \Bigg(\max_{k\in[\min\{j,k^*\}]} \{c_{jk}^t - c_j^r\}\Bigg)\Bigg\}\Bigg\},$$

where c_j^r, c_{jk}^t, c_1^p, *and* c_{1k}^s *are defined through* (9e)–(9h).

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Theorem 4 (see Online Appendix K for a proof) enables us to reduce the $(J+1)!$ many constraints (9c) in the reformulation of (DRNS) to only $\frac{1}{2}J(J+1)+1$ many, thanks to the closed-form solution. This leads to the MILP reformulation of (DRNS) in Proposition 4 (see Online Appendix L for a proof).

Proposition 4. *Under Assumption 1 and Structure 1, the (DRNS) model yields the same optimal objective value as the following MILP:*

$$\begin{aligned} Z_1^\star := \min\ & \theta + c_1^y y_1^L + g_1(y_1^L)\lambda_1 + \sum_{\ell=1}^{y_1^U - y_1^L} (c_1^y v_{1\ell} + \delta_{1\ell}\nu_{1\ell}) \\ & + \sum_{j=1}^{J}\Bigg[\sum_{q=1}^{Q}\mu_{jq}\rho_{jq} + c_j^w w_j^L + f_j(w_j^L)\gamma_j \\ & + \sum_{\ell=1}^{w_j^U - w_j^L}(c_j^w u_{j\ell} + \Delta_{j\ell}\varphi_{j\ell})\Bigg] \\ \text{s.t.}\ & (8a)-(8f),\ (9d), \\ & \theta \ge \phi_1^e + \sum_{j=1}^{J}(\zeta_j^e + \eta_j^e), \\ & \theta \ge \sum_{j=1}^{J}(\zeta_j^e + \eta_j^e) + \phi_{1k^*}^x + (\zeta_{j^*k^*}^x + \eta_{j^*k^*}^x) \\ & \quad - (\zeta_{j^*}^e + \eta_{j^*}^e) + \sum_{j\in[J]\setminus\{j^*\}} \chi_{jk^*},\ \forall j^*\in[J],\ \forall k^*\in[j^*], \\ & \chi_{jk^*} \ge (\zeta_{jk}^x + \eta_{jk}^x) - (\zeta_j^e + \eta_j^e),\ \forall j\in[J],\ \forall k^*\in[J], \\ & \qquad \forall k\in[\min\{j,k^*\}], \end{aligned} \tag{13a}$$

$$\left.\begin{aligned} & \zeta_j^e \ge -\gamma_j w_j^L - \sum_{\ell=1}^{w_j^U - w_j^L}\varphi_{j\ell},\ \zeta_j^e \ge 0, \\ & \eta_j^e \ge -\sum_{q=1}^{Q}\tilde{d}_j^q\rho_{jq},\ \forall \tilde{d}_j\in[d_j^L, d_j^U]_{\mathbb{Z}}, \end{aligned}\right\}\ \forall j\in[J], \tag{13b}$$

$$\left.\begin{aligned} & \zeta_{jk}^x \ge (-c_k^x - \gamma_j)w_j^L - \sum_{\ell=1}^{w_j^U - w_j^L}(c_k^x u_{j\ell} + \varphi_{j\ell}),\ \zeta_{jk}^x \ge 0, \\ & \eta_{jk}^x \ge c_k^x\tilde{d}_j - \sum_{q=1}^{Q}\tilde{d}_j^q\rho_{jq},\ \forall \tilde{d}_j\in[d_j^L, d_j^U]_{\mathbb{Z}}, \\ & \qquad \forall j\in[J],\ \forall k\in[j], \end{aligned}\right\} \tag{13c}$$

$$\begin{aligned} & \phi_1^e \ge -\lambda_1 y_1^L - \sum_{\ell=1}^{y_1^U - y_1^L}\nu_{1\ell},\ \phi_1^e \ge 0, \\ & \phi_{1k}^x \ge (-c_k^x - \lambda_1)y_1^L - \sum_{\ell=1}^{y_1^U - y_1^L}(c_k^x v_{1\ell} + \nu_{1\ell}),\ \phi_{1k}^x \ge 0, \\ & \qquad \forall k\in[J]. \end{aligned}$$

A special case of Structure 1 is when there is no nurse float pool, which is equivalent to assigning all units to a pool with no nurses. We call this Structure 0, where $y_1^L = y_1^U = 0$. A MILP reformulation of (DRNS) under Structure 0 follows from Proposition 4:

$$\begin{aligned} Z_0^\star := \min\ & \theta + \sum_{j=1}^{J}\Bigg[\sum_{q=1}^{Q}\mu_{jq}\rho_{jq} + c_j^{\mathrm{w}} w_j^{\mathrm{L}} + f_j(w_j^{\mathrm{L}})\gamma_j \\ & + \sum_{\ell=1}^{w_j^{\mathrm{U}} - w_j^{\mathrm{L}}}(c_j^{\mathrm{w}} u_{j\ell} + \Delta_{j\ell}\varphi_{j\ell})\Bigg] \\ \text{s.t.}\ & (8a),\ (8b),\ (8e),\ (9d),\ (13a)–(13c), \\ & \theta \ge \sum_{j=1}^{J}(\zeta_j^{\mathrm{e}} + \eta_j^{\mathrm{e}}), \\ & \theta \ge \sum_{j=1}^{J}(\zeta_j^{\mathrm{e}} + \eta_j^{\mathrm{e}}) + (\zeta_{j^*k^*}^{\mathrm{x}} + \eta_{j^*k^*}^{\mathrm{x}}) - (\zeta_{j^*}^{\mathrm{e}} + \eta_{j^*}^{\mathrm{e}}) \\ & \quad + \sum_{j\in[J]\setminus\{j^*\}}\chi_{jk^*},\ \forall j^*\in[J],\ \forall k^*\in[j^*]. \end{aligned}$$

We notice that, whenever $y_1^{\mathrm{L}} = 0$, any feasible nurse staffing levels under Structure 0 are also feasible to (DRNS) under Structure 1. It follows that $Z_1^\star \le Z_0^\star$. In addition, because Structure 1 provides the most flexibility and Structure 0 has zero flexibility, we interpret the difference $Z_0^\star - Z_1^\star$ as the (maximum) value of flexibility.

4.2. Disjoint Pools

Under Structure D, we denote by $\mathcal{H}_{\mathrm{D}}$ the feasible region $\mathcal{H}$ of the integer program (7). Intuitively, as the nurse pools are disjoint, $\mathcal{H}_{\mathrm{D}}$ becomes separable in index i, that is, separable among the nurse pools and the units pertaining to each pool. Hence, we obtain $\mathrm{conv}(\mathcal{H}_{\mathrm{D}})$ by convexifying the projection of $\mathcal{H}_{\mathrm{D}}$ in each pool and then taking their Cartesian product. It follows that, once again, the convex maximization problem $\max_{(\alpha,\beta)} F(\alpha,\beta)$ admits a closed-form solution and (DRNS) can be recast as a MILP. In particular, we reduce the exponentially many constraints (9c) to only $0.5J(J+1)+I$ many. We summarize these results in Proposition 5 (see Online Appendix M for a proof).

Proposition 5. *Under Structure D, it holds that* $\mathrm{conv}(\mathcal{H}_{\mathrm{D}}) = \{(t,s,r,p) : (7b), (10)\}$. *In addition, for fixed* $(u,v,\gamma,\lambda,\rho)$, *the optimal value of* $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ *equals*

$$\sum_{i=1}^{I}\Bigg[\max\Bigg\{c_i^p + \sum_{j\in P_i} c_j^r,\ \max_{\substack{j^*\in P_i,\\ k^*\in[j^*]}}\Bigg\{\sum_{j\in P_i} c_j^r + c_{ik^*}^s + (c_{j^*k^*}^t - c_{j^*}^r) + \sum_{j\in P_i\setminus\{j^*\}}\Bigg(\max_{k\in[\min\{j,k^*\}]}\{c_{jk}^t - c_j^r\}\Bigg)\Bigg\}\Bigg\}\Bigg],$$

where c_j^r, c_{jk}^t, c_i^p, *and* c_{ik}^s *are defined through* (9e)–(9h).

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Furthermore, under Assumption 1, the (DRNS) model yields the same optimal objective value as the following MILP:

$$Z_D^\star := \min \sum_{i=1}^{I} \left(\theta_i + c_i^y y_i^L + g_i(y_i^L)\lambda_i + \sum_{\ell=1}^{y_i^U - y_i^L} (c_i^y v_{i\ell} + \delta_{i\ell} \nu_{i\ell}) \right)$$
$$+ \sum_{j=1}^{J} \left[\sum_{q=1}^{Q} \mu_{jq}\rho_{jq} + c_j^w w_j^L + f_j(w_j^L)\gamma_j \right.$$
$$\left. + \sum_{\ell=1}^{w_j^U - w_j^L} (c_j^w u_{j\ell} + \Delta_{j\ell}\varphi_{j\ell}) \right]$$

s.t. (8a)–(8f), (9d), (13a)–(13c),

$$\theta_i \geq \phi_i^e + \sum_{j \in P_i} (\zeta_j^e + \eta_j^e), \ \forall i \in [I],$$

$$\theta_i \geq \sum_{j \in P_i} (\zeta_j^e + \eta_j^e) + \phi_{ik^*}^x + (\zeta_{j^*k^*}^x + \eta_{j^*k^*}^x) - (\zeta_{j^*}^e + \eta_{j^*}^e)$$
$$+ \sum_{j \in P_i \setminus \{j^*\}} \chi_{jk^*}, \quad \forall i \in [I], \forall j^* \in P_i, \forall k^* \in [j^*],$$

$$\phi_i^e \geq -\lambda_i y_i^L - \sum_{\ell=1}^{y_i^U - y_i^L} \nu_{i\ell}, \ \phi_i^e \geq 0, \ \forall i \in [I], \tag{14a}$$

$$\phi_{ik}^x \geq (-c_k^x - \lambda_i) y_i^L - \sum_{\ell=1}^{y_i^U - y_i^L} (c_k^x v_{i\ell} + \nu_{i\ell}), \ \phi_{ik}^x \geq 0,$$
$$\forall i \in [I], \forall k \in [J(i)]. \tag{14b}$$

4.3. Chained Pools

Under Structure C, the inequalities (10) can strengthen the mixed-integer set $\mathcal{H}$ but no longer describe $\text{conv}(\mathcal{H})$. Nevertheless, we adopt an alternative approach to recast the convex maximization problem $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ as a linear program. We start by noticing that, by renumbering the units if needed, the chained pools can be made $P_i = \{i, i+1\}$ for all $i \in [I-1]$ and $P_I = \{I, 1\}$. For ease of exposition, we denote by $\sigma(i)$ the original index of unit i before renumbering. We define an order $\prec$ on $[I]$ such that $i \prec j$ if and only if $\sigma(i) < \sigma(j)$, and we say $i \preceq j$ if $i \prec j$ or $i = j$. In addition, we let $i \Upsilon j$ represent the index of higher order between units i and j, that is, $i \Upsilon j = j$ if $i \preceq j$ and $i \Upsilon j = i$ otherwise. We create a dummy unit $I+1$, which is a duplicate of unit 1, and refer to these two units interchangeably.

Example 1. Consider the chained pools in Figure 3(a) with 3 units. After renumbering (see Figure 3(b)), we have $\sigma(1) = 1$, $\sigma(2) = 3$, and $\sigma(3) = 2$. In addition, $1 \prec 3 \prec 2$ and $2 \Upsilon 3 = 2$.

Next, because $P_i = \{i, i+1\}$ for all $i \in [I]$, variables s_{ik} in the integer program (7) depend on $t_{[i]}$ and $t_{[i+1]}$ only, where $t_{[i]} := \{t_{ik} : k \preceq i\}$. Indeed, the definition of s_{ik} implies

$$s_{ik} = h_k(t_{[i]}, t_{[i+1]})$$
$$:= \begin{cases} 1 & \text{if } \max\{t_{ik}, t_{(i+1)k}\} = 1, \ t_{i\ell} = 0 \text{ for all } \ell \succ k, \\ & \text{and } t_{(i+1)\ell} = 0 \text{ for all } \ell \succ k, \\ 0 & \text{otherwise.} \end{cases}$$

Plugging these representations into formulation (7) yields a reformulation of $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ based on variables t only:

$$\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta) = \sum_{i=1}^{I} (c_i^r + c_i^p) + \sum_{i=1}^{I} \left[\sum_{k \preceq i} (c_{ik}^t - c_i^r) t_{ik} \right.$$
$$\left. + \sum_{k \preceq i \Upsilon (i+1)} (c_{ik}^s - c_i^p) h_k(t_{[i]}, t_{[i+1]}) \right]. \tag{15}$$

The reformulation (15) decomposes objective function based on index $i \in [I]$ and enables us to solve $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ by a dynamic program (DP), that is, we sequentially optimize $t_{[1]}, t_{[2]}, \ldots, t_{[I]}$. To this end, we define the state of the DP in stage 1 as $t_{[1]} \equiv t_{11} \in \mathcal{B}_1$ and the states in stage i as $(t_{[1]}, t_{[i]}) \in \mathcal{B}_1 \times \mathcal{B}_i$ for all $i \in [2, I]_{\mathbb{Z}}$, where $\mathcal{B}_i := \{t_{[i]} \in \mathbb{B}^{\sigma(i)} : \sum_{k \preceq i} t_{ik} \leq 1\}$ represents the feasible region of $t_{[i]}$ described by constraints (6a). In addition, we formulate the DP as $\max_{(t_{[1]}, t_{[I]}) \in \mathcal{B}_1 \times \mathcal{B}_I} \{V_I(t_{[1]}, t_{[I]}) + \sum_{k^1 (1 \Upsilon I)} (c_{Ik}^{\text{s}} - c_I^{\text{p}}) h_k(t_{[I]}, t_{[1]})\}$ where the value functions $V_i(\cdot)$ are recursively defined through

$$V_1(t_{[1]}) := \sum_{i=1}^{I} (c_i^r + c_i^p) + (c_{11}^t - c_1^r) t_{11} \text{ and}$$

$$V_i(t_{[1]}, t_{[i]}) := \max_{t_{[i-1]} \in \mathcal{B}_{i-1}} \left\{ V_{i-1}(t_{[1]}, t_{[i-1]}) + \sum_{k \preceq i} (c_{ik}^{\text{t}} - c_i^{\text{r}}) t_{ik} \right.$$
$$\left. + \sum_{k \preceq (i-1) \Upsilon i} (c_{(i-1)k}^{\text{s}} - c_{i-1}^{\text{p}}) h_k(t_{[i-1]}, t_{[i]}) \right\}$$

for all $i \in [2, I]_{\mathbb{Z}}$ and $(t_{[1]}, t_{[i]}) \in \mathcal{B}_1 \times \mathcal{B}_i$. For all $i \in [I]$, the value function $V_i(t_{[1]}, t_{[i]})$ represents the "cumulative reward" up to stage i, that is, the terms in (15) that involve $t_{[1]}, \ldots, t_{[i]}$ only. We note that, as $t_{[1]}$ is involved in the final-stage reward, the DP stores the value of $t_{[1]}$ in the state throughout stages $2, \ldots, I$.

We further interpret the DP as a longest-path problem on an acyclic directed network $(\mathcal{N}, \mathcal{A})$. Specifically, the set of nodes $\mathcal{N}$ consists of I layers, denoted by $\{N_i\}_{i=1}^{I}$. For all $i \in [I]$, layer i consists of the states of the DP in stage i, that is, $\mathcal{N}_1 = \mathcal{B}_1 = \{0, 1\}$ and $\mathcal{N}_i = \mathcal{B}_1 \times \mathcal{B}_i$ for all $i \in [2, I]_{\mathbb{Z}}$. In addition, $\mathcal{A}$ consists of arcs that connect two nodes in neighboring layers, as long as the two nodes share a common $t_{[1]}$ value, that is, $\mathcal{A} = \{[t_{[1]}, (t_{[1]}, t_{[2]})] : t_{[1]} \in \mathcal{B}_1, t_{[2]} \in \mathcal{B}_2\} \cup \{[(t_{[1]}, t_{[i-1]}), (t_{[1]}, t_{[i]})] : t_{[i-1]} \in \mathcal{B}_{i-1}, t_{[i]} \in \mathcal{B}_i, \ \forall i \in [3, I]_{\mathbb{Z}}\}$. Finally, we incorporate into $\mathcal{N}$ a

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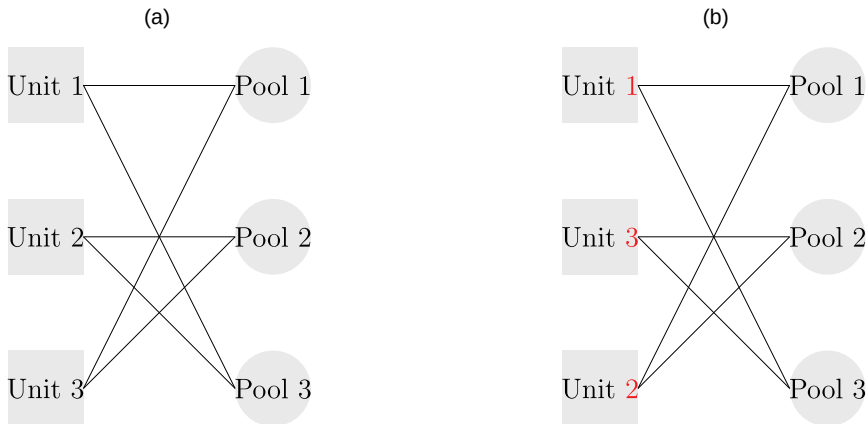
Figure 3. (Color online) Renumbering Units in Structure C



Notes. (a) Before. (b) After.

starting node $\mathtt{S}$ and a terminal node $\mathtt{T}$, and into $\mathcal{A}$ arcs from $\mathtt{S}$ to all nodes in $\mathcal{N}_1$ and from all nodes in $\mathcal{N}_I$ to $\mathtt{T}$. Then, the DP is equivalent to the longest-path problem from $\mathtt{S}$ to $\mathtt{T}$ on $(\mathcal{N},\mathcal{A})$. We formally state this result in the following theorem. For notation brevity, we say $t_{[i]} = e_{ik}$ if $t_{ik} = 1$ for a $k \preceq i$, while $t_{i\ell} = 0$ for all $\ell \preceq i$ and $\ell \neq k$; and we say $t_{[i]} = e_{i0}$ if $t_{ik} = 0$ for all $k \preceq i$. In Figure 4, we depict $(\mathcal{N},\mathcal{A})$ for Example 1.

Theorem 5. *Define $\{c_{[m,n]} : [m,n] \in \mathcal{A}\}$, the length of the arcs in network $(\mathcal{N},\mathcal{A})$, such that*

$$c_{[\mathtt{S},t_{[1]}]} = \sum_{i=1}^{I}(c_i^r + c_i^p) + (c_{11}^t - c_1^r)t_{11} \quad \forall t_{[1]} \in \mathcal{B}_1,$$

$$c_{[t_{[1]},(t_{[1]},e_{2\ell})]} = \begin{cases} (c_{11}^s - c_1^p)t_{11} & \text{if } \ell = 0 \\ (c_{2k}^t - c_2^r) + (c_{1k}^s - c_1^p) & \text{if } \ell \neq 0 \end{cases} \quad \forall t_{[1]} \in \mathcal{B}_1,\ \forall \ell \preceq 2,$$

$$c_{[(t_{[1]},e_{(i-1)k}),(t_{[1]},e_{i\ell})]} = \begin{cases} 0 & \text{if } k = 0,\ \ell = 0 \\ c_{(i-1)k}^s - c_{i-1}^p, & \text{if } k \neq 0,\ \ell = 0 \\ (c_{i\ell}^t - c_i^r) + (c_{(i-1)(k\curlyvee\ell)}^s - c_{i-1}^p), & \text{if } \ell \neq 0 \end{cases}$$

$$\forall t_{[1]} \in \mathcal{B}_1,\ \forall i \in [3,I]_{\mathbb{Z}},\ \forall k \preceq i-1,\ \forall \ell \preceq i,$$

$$\text{and } c_{[(t_{[1]},e_{Ik}),\mathtt{T}]} = \begin{cases} (c_{I1}^s - c_I^p)t_{11} & \text{if } k = 0 \\ c_{Ik}^s - c_I^p & \text{if } k \neq 0 \end{cases} \quad \forall t_{[1]} \in \mathcal{B}_1,\ \forall k \preceq I.$$

Then, $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ equals the length of the longest $\mathtt{S}$–$\mathtt{T}$ path on $(\mathcal{N},\mathcal{A})$, that is,

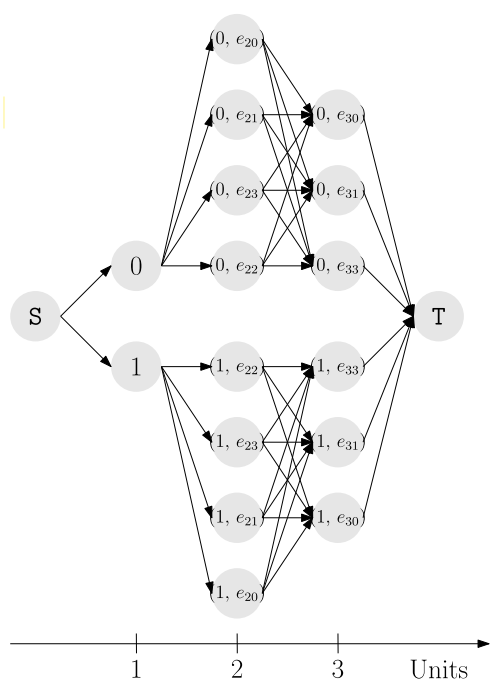
$$\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta) = \max_{x\in[0,1]^{\mathcal{A}}} \sum_{[m,n]\in\mathcal{A}} c_{[m,n]}x_{[m,n]}$$

$$\text{s.t.} \sum_{n:[m,n]\in\mathcal{A}} x_{[m,n]} - \sum_{n:[n,m]\in\mathcal{A}} x_{[n,m]} = \begin{cases} 1 & \text{if } m = \mathtt{S} \\ 0 & \text{if } m \neq \mathtt{S},\mathtt{T} \\ -1 & \text{if } m = \mathtt{T} \end{cases}, \quad \forall m \in \mathcal{N}.$$

We note that $(\mathcal{N},\mathcal{A})$ is acyclic and it consists of $O(I^2)$ nodes and $O(I^3)$ arcs. Hence, $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ can be obtained in time *polynomial of the problem input*. Accordingly, we replace the exponentially many constraints (9c) in the formulation of (DRNS) with only $O(I^3)$ many linear ones. This yields the MILP reformulation in Proposition 6 (see Online Appendix O for a proof).

Proposition 6. *Under Structure C and Assumption 1, the (DRNS) model yields the same optimal objective value as*

Figure 4. Longest-Path Problem on an Acyclic Directed Network



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the following MILP:

$$\min\ \theta + \sum_{i=1}^{I}(c_i^y y_i^L + g_i(y_i^L)\lambda_i + \sum_{\ell=1}^{y_i^U - y_i^L}(c_i^y v_{i\ell} + \delta_{i\ell}\nu_{i\ell}))$$

$$+ \sum_{j=1}^{I}\left[\sum_{q=1}^{Q}\mu_{jq}\rho_{jq} + c_j^w w_j^L + f_j(w_j^L)\gamma_j + \sum_{\ell=1}^{w_j^U - w_j^L}(c_j^w u_{j\ell} + \Delta_{j\ell}\varphi_{j\ell})\right]$$

$$\text{s.t.}\quad (8a)\text{–}(8f),\ (9d),\ (13b)\text{–}(13c),\ (14a)\text{–}(14b),$$

$$\theta \geq \sum_{i=1}^{I}(c_i^r + c_i^p) + \pi_{\mathrm{S}} - \pi_{\mathrm{T}},$$

$$\pi_{\mathrm{S}} - \pi_{t_{[1]}} \geq ((\zeta_{11}^x + \eta_{11}^x) - (\zeta_1^e + \eta_1^e))t_{11} \qquad \forall t_{[1]} \in \mathcal{B}_1,$$

$$\pi_{t_{[1]}} - \pi_{(t_{[1]}, e_{2\ell})} \geq \begin{cases} (\phi_{11}^x - \phi_1^e)t_{11} & \text{if } \ell = 0 \\ (\zeta_{2\ell}^x + \eta_{2\ell}^x) - (\zeta_2^e + \eta_2^e) & \\ \quad + (\phi_{1\ell}^x - \phi_1^e) & \text{if } \ell \neq 0 \end{cases} \qquad \forall t_{[1]} \in \mathcal{B}_1,\ \forall \ell \preceq 2,$$

$$\pi_{(t_{[1]}, e_{(i-1)k})} - \pi_{(t_{[1]}, e_{i\ell})} \geq \begin{cases} 0, & \text{if } k = 0, \ell = 0 \\ \phi_{(i-1)k}^x - \phi_{i-1}^e, & \text{if } k \neq 0, \ell = 0 \\ (\zeta_{i\ell}^x + \eta_{i\ell}^x) - (\zeta_i^e + \eta_i^e) & \text{if } \ell \neq 0 \\ \quad + (\phi_{(i-1)(k\curlyvee\ell)}^x - \phi_{i-1}^e), & \end{cases}$$

$$\forall t_{[1]} \in \mathcal{B}_1,\ \forall i \in [3, I]_{\mathbb{Z}},\ \forall k \preceq i-1,\ \forall \ell \preceq i,$$

$$\pi_{(t_{11}, e_{Ik})} - \pi_{\mathrm{T}} \geq \begin{cases} (\phi_{I1}^x - \phi_I^e)t_{11} & \text{if } k = 0 \\ \phi_{Ik}^x - \phi_I^e & \text{if } k \neq 0 \end{cases} \quad \forall t_{[1]} \in \mathcal{B}_1,\ \forall k \preceq I.$$

5. Numerical Experiments

We report numerical experiments on the (DRNS) model, using data from our collaborating hospital (see Section 5.1). We summarize our main findings as follows:

1. Under the practical nurse pool structures as introduced in Section 4, the MILP reformulations of (DRNS) lead to a significant speed-up over the separation algorithm (see Section 5.2).

2. Allowing the absenteeism rate to vary with respect to the staffing level, as opposed to treating it as a constant, delivers better out-of-sample performance (see Section 5.3).

3. Modeling nurse absenteeism, as opposed to overlooking it, produces significantly better staffing decisions and out-of-sample performance (see Section 5.4).

4. Incorporating higher moments of random show-up nurses leads to an approximate solution which could have better out-of-sample performance compared with (DRNS) (see Section 5.5).

In all experiments, we solve optimization models by GUROBI 9.0.1 via Python 2.7 on a personal laptop with an Intel® Core™ i7-4850HQ CPU@2.3 GHz and 16 GB RAM.

5.1. Data

Our test instances are based on five units, abbreviated as CVC5, 4B, 4C, 7A1, and 7B in our collaborating hospital. We obtain the nurse demand and attendance entries of these five units from July 2010 to June 2014. To calibrate the ambiguity set $\mathcal{D}(w, y)$, we randomly split the data into a training data set (taking up 80% of all data entries) and a testing data set (20%). We set $Q := 2$, that is, we consider mean value μ_{j1} and standard deviation sd_j of the nurse demand, and we set $(d_j^{\mathrm{L}}, d_j^{\mathrm{U}}, \mu_{j1}, \mathrm{sd}_j)$ as the empirical min, max, mean, and standard deviation of the training data, respectively. Likewise, we set $(w_j^{\mathrm{L}}, w_j^{\mathrm{U}}, f_j(w_j))$ as the empirical min, max, and mean of the training data. For pool nurses, we assume a constant absence rate such that, for all $i \in [I]$, $g_i(y_i) := (1 - A_i^{\mathrm{p}})y_i$, where A_i^{p} is randomly extracted from the interval $[0, 0.02]$. Additionally, we set $y_i^{\mathrm{L}} := 0$ and $y_i^{\mathrm{U}} := \sum_{j \in P_i}(d_j^{\mathrm{U}} - w_j^{\mathrm{L}})$. We summarize the parameters of $\mathcal{D}(w, y)$ in Table 1.

5.2. Computational Efficacy

We evaluate the computational efficacy of three approaches on solving (DRNS) under general and special pool structures: the separation algorithm described in Algorithm 1 (denoted by Sep), the separation algorithm with valid inequalities (10) (denoted by $\mathrm{Sep}_{\mathrm{VI}}$), and the MILP reformulations derived in Section 4 in case of special pool structures (denoted by MILP). To evaluate the computational efficacy, we duplicate the five-unit system to create 10- and 50-unit systems. For each system, we consider four different pool structures, including general pool and Structures 1, D, and C. We report the computing time (in wall-clock seconds) in Table 2. From this table, we observe that valid inequalities (10) significantly improve the computational efficacy of the separation algorithm. For example, the computing time of $\mathrm{Sep}_{\mathrm{VI}}$ is a small fraction of that of Sep

Table 1. Parameters of $\mathcal{D}(w, y)$ in the Five-Unit System

	Demand				Decision-dependent nurse show-up		
Unit j	d_j^{L}	d_j^{U}	μ_{j1}	sd_j	w_j^{L}	w_j^{U}	$\{(w_j, f_j(w_j)) : w_j \in [w_j^{\mathrm{L}}, w_j^{\mathrm{U}}]_{\mathbb{Z}}\}$
1	4	12	10.39	1.75	9	12	{(9, 7), (10, 7.55), (11, 7.57), (12, 7.85)}
2	6	14	12.91	1.34	10	13	{(10, 8.5), (11, 8.89), (12, 9), (13, 9)}
3	8	18	16.60	1.55	14	17	{(14, 11.64), (15, 11.68), (16, 11.86), (17, 12)}
4	5	14	12.08	1.48	8	13	{(8, 7), (9, 7.5), (10, 8), (11, 8.75), (12, 8.82), (13, 9)}
5	6	16	14.58	1.42	12	16	{(12, 10), (13, 10.55), (14, 11), (15, 10.83), (16, 11)}

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Table 2. Wall-Clock Seconds Taken to Solve (DRNS) by Separation Algorithm (Sep), Separation Algorithm with Valid Inequalities (Sep_{VI}), and MILP Reformulations in Section 4

	General[a]		Structure 1			Structure D[b]			Structure C		
J	Sep	Sep_{VI}	Sep	Sep_{VI}	MILP	Sep	Sep_{VI}	MILP	Sep	Sep_{VI}	MILP
5	0.41	0.27	0.22	0.20	0.18	0.23	0.22	0.16	0.37	0.26	0.25
10	4.02	0.93	0.57	0.46	0.52	0.71	0.40	0.22	23.15	0.83	0.67
50	285.19	50.93	81.27	15.72	17.20	165.85	44.62	8.22	Limit[c]	3,187.35	1,078.28

[a]$P_1 = \{j \in [J] : j \text{ is even}\}$, $P_2 = \{j \in [J] : j \text{ is odd}\}$, $P_3 = \{j \in [J] : j \le J/2\}$, and $P_4 = \{j \in [J] : j > J/2\}$.
[b]$P_i = \{4i-3, 4i-2, 4i-1, 4i\}$, $\forall i \in [\lfloor J/4 \rfloor]$, and $P_{\lfloor J/4 \rfloor + 1} = \{4\lfloor J/4 \rfloor + 1, \ldots, J\}$.
[c]Time limit = two hours.

in most instances, and Sep_{VI} solves the most challenging instance ($J = 50$ under Structure C) to optimum within the time limit, whereas Sep cannot. In addition, MILP prevails under special pool structures. For example, it solves all instances to global optimum in 18 minutes.

5.3. Comparing Constant and Varying Absenteeism Rates

When characterizing the ambiguity set $\mathcal{D}(w, y)$, we can model f_j and g_i simply as linear functions of the staffing levels (w, y). Linear f_j and g_i imply a constant nurse show-up rate (i.e., the expected number of show-up nurses divided by the staffing levels), which is independent of the staffing decisions. This gives rise to a DRO^{exo} model with constant absenteeism (or show-up) rates. That is, the DRO^{exo} approach assumes that the expected number of the nurses showing up is a linear function of the staffing level with a constant slope. This is in contrast to DRNS, wherein the absenteeism rates may vary pertaining to the staffing level (see, e.g., Figure 1 (right)). In this section, we evaluate the out-of-sample performance of DRO^{exo} and DRNS solutions to compare these two modeling choices.

We compute the constant show-up rates in DRO^{exo} by averaging across all nurse attendance records in the data. In addition, we present the formulation of DRO^{exo} in Online Appendix P. We report the out-of-sample average staffing costs of DRNS and DRO^{exo} in Table 3. From this table, we observe that DRNS leads to a lower staffing cost on average and tends to need fewer temporary nurses than DRO^{exo}. Under a general pool structure, as well as under Structures 1 and C, the gap in out-of-sample performance exceeds 25%. This demonstrates the value of modeling varying absenteeism rates. To examine this observation further, we compare the optimal staffing levels (w^*, y^*) of DRNS and DRO^{exo} in Table 4. From this table, we observe that DRNS staffs more pool nurses and assigns fewer unit nurses. This is because, in our instances, the nurses become less likely to show up as the staffing level increases (see Figure 1 (right)). The increasing absenteeism rate discourages DRNS from staffing more unit nurses. In contrast, the DRO^{exo} approach assumes a constant absenteeism rate and overstaffs unit nurses, resulting in out-of-sample disappointment.

5.4. The Value of Modeling Absenteeism

As discussed in Section 1, modeling nurse absenteeism incurs decision-dependent uncertainty and computational challenges. It is hence worth examining what (DRNS) buys us, that is, the value of modeling absenteeism. To this end, we generate 400 instances by tuning parameters of the five-unit system under Structure 1. Specifically, we (i) generate random c_j^{x} from the interval $[450, 1500]$, (ii) perturb μ_{j1} in Table 1 by adding a random noise $\tilde{\alpha}_j$ supported on the interval $[-1, 1]$, and (iii) generate random sd_j from the interval $[1, 2]$. In addition, we consider a variant of (DRNS) that overlooks the nurse absenteeism, in which we assume that all assigned nurses show up. Then, we compare the out-of-sample performance of the optimal nurse staffing decisions produced by (DRNS) and that produced by overlooking absenteeism. Fixing the nurse staffing levels as in a (DRNS) optimal solution (w^*, y^*), we

Table 3. Out-of-Sample Comparison Between DRO^{exo} and DRNS

	Average staffing cost			Average number of temp nurses	
	DRO^{exo}	DRNS	GAP (%)[a]	DRO^{exo}	DRNS
Structure 0	35,030	33,996	3.04	17.91	18.87
Structure 1	31,826	25,268	25.95	12.92	4.93
Structure D	32,597	31,051	4.98	14.08	14.12
Structure C	32,859	25,706	27.83	13.94	4.95
General	31,908	24,912	28.08	12.99	4.15

[a]GAP= $(Z^{\text{DRO(exo)}} - Z^{\text{DRNS}})/Z^{\text{DRNS}} \times 100\%$, where $Z^{\text{DRO(exo)}}$ and Z^{DRNS} are average staffing costs of DRO^{exo} and (DRNS) in the out-of-sample simulation, respectively.

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Table 4. Optimal Staffing Levels Produced by DRO^{exo} and DRNS

	DRO^{exo}		DRNS	
	w^*	y^*	w^*	y^*
Structure 0	10, 13, 17, 12, 16	0	10,11, 14, 11, 14	0
Structure 1	9, 13, 17, 12, 16	5	9, 10, 14, 8, 12	17
Structure D	9, 13, 17, 12, 16	4	9, 11, 14, 11, 14	5
Structure C	9, 13, 17, 12, 16	0, 1, 1, 0, 3	9, 10, 14, 8, 12	3, 4, 3, 5, 3
General	9, 13, 17, 12, 16	4, 1	9, 10, 14, 8, 12	9, 9

generate a large number of scenarios for nurse demand and absenteeism. Exposing (w^*, y^*) under these scenarios produces an out-of-sample estimate of the average staffing cost with absenteeism, which we denote by Z_{abs}. Using the same set of scenarios, we examine the optimal solution produced by overlooking absenteeism and obtain an out-of-sample average cost without absenteeism, denoted by $Z_{\text{w/o}}$. Using the same out-of-sample procedures, we compute the average number of temporary nurses hired when considering absenteeism (denoted by x_{abs}) and when overlooking it (denoted by $x_{\text{w/o}}$).

We depict the values of $Z_{\text{w/o}}$ (x-coordinate) and Z_{abs} (y-coordinate) obtained in 400 replications in Figure 5(a). From this figure, we observe that most dots are below the 45-degree line, indicating that $Z_{\text{w/o}} - Z_{\text{abs}} > 0$, that is, modeling nurse absenteeism yields nurse staffing levels with better out-of-sample performance. In addition, we group the dots based on the relative value of flexibility $\text{OVG} := (Z_0^\star - Z_1^\star)/Z_0^\star \times 100\%$. From Figure 5(a), we observe that the difference $Z_{\text{w/o}} - Z_{\text{abs}}$ shows an increasing trend as OVG increases. That is, modeling nurse absenteeism becomes more valuable as the value of flexibility increases. This makes sense because when a unit is short of supply because of nurse absenteeism, making it up with pool nurses is less expensive than doing so with temporary nurses. As a result, setting up nurse pools can effectively mitigate the impacts of nurse absenteeism. In Figure 5(b), we depict the values of $x_{\text{w/o}}$ and x_{abs} obtained in the 400 replications and make similar observations.

5.5. Higher-Order Moments of Random Show-Up Nurses

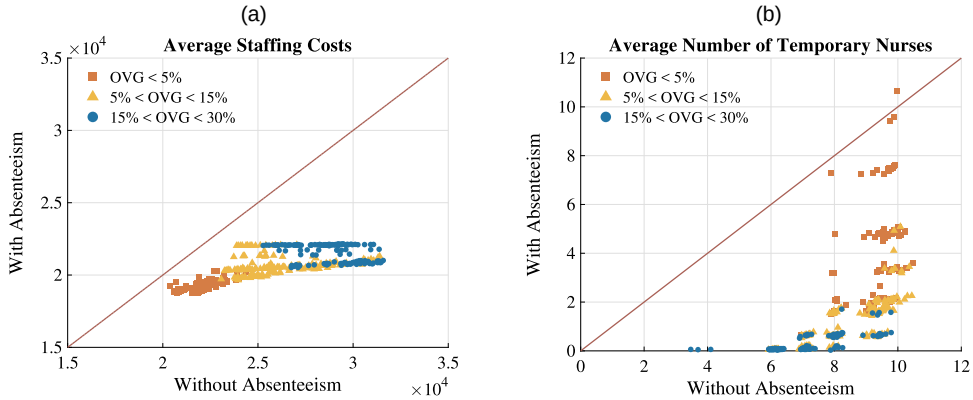
So far, the ambiguity set $\mathcal{D}(w,y)$ only considers the first-order moments (or mean values) for the random show-up nurses $(\tilde{w}, \tilde{y})$. In this section, we incorporate higher-order (up to the Q-th order, with $Q \geq 2$) moments of $(\tilde{w}, \tilde{y})$ and extend $\mathcal{D}(w,y)$ to be

$$\mathcal{D}(w,y) = \{\mathbb{P} \in \mathcal{P}(\Xi) : \mathbb{E}_{\mathbb{P}}[\tilde{d}_j^q] = \mu_{jq}, \quad \mathbb{E}_{\mathbb{P}}[\tilde{w}_j^q] = f_{jq}(w_j), \quad \forall j \in [J], \ \forall q \in [Q], \tag{16a}$$

$$\mathbb{E}_{\mathbb{P}}[\tilde{y}_i^q] = g_{iq}(y_i), \quad \forall i \in [I], \ \forall q \in [Q]\}. \tag{16b}$$

As noted in Section 2, the extended $\mathcal{D}(w,y)$ can be calibrated through the nurse demand data and attendance records, and we denote as DRNS$^{(Q)}$ the ensuing staffing model using the extended $\mathcal{D}(w,y)$. We recall that the support set Ξ depends on the staffing levels w and y. This makes DRNS$^{(Q)}$ intractable, as its reformulation boils down to a set of integer programs with polynomial objective functions and changing feasible regions pertaining to (w, y). Fortunately, relaxing (w,y) to their

Figure 5. (Color online) An Out-of-Sample Performance Comparison of Considering vs. Overlooking Nurse Absenteeism



Note. Note that $\text{OVG} := (Z_0^\star - Z_1^\star)/Z_0^\star \times 100\%$, where $Z_0^\star$ (respectively, $Z_1^\star$) is the optimal value of (DRNS) with no nurse pools (respectively, under Structure 1).

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upper bounds $(w^{\mathrm{U}}, y^{\mathrm{U}})$ renders the feasible region decision-independent. In this case, we can solve these integer programs by enumerating a linear number of staffing levels. This produces a conservative and tractable approximation for DRNS$^{(Q)}$:

$$\min_{\gamma,\lambda,\rho^{w,y}} \max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta) + \sum_{i=1}^{I}\left[c_i^{\mathrm{y}} y_i + \sum_{q=1}^{Q} g_{iq}(y_i)\lambda_{iq}\right] + \sum_{j=1}^{J}\left[c_j^{\mathrm{w}} w_j + \sum_{q=1}^{Q}\mu_{jq}\rho_{jq} + \sum_{q=1}^{Q} f_{jq}(w_j)\gamma_{jq}\right] \tag{17a}$$

$$\text{s.t.} \quad (3b)\text{–}(3c), \tag{17b}$$

where $\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta)$ is recast as a MILP:

$$\max_{(\alpha,\beta)\in\Lambda} F(\alpha,\beta) := \max_{(t,s,r,p)\in\mathcal{H}} \sum_{j=1}^{J}\left(c_j^{\mathrm{r}} r_j + \sum_{k=1}^{j} c_{jk}^{\mathrm{t}} t_{jk}\right) + \sum_{i=1}^{I}\left(c_i^{\mathrm{p}} p_i + \sum_{k=1}^{J(i)} c_{ik}^{\mathrm{s}} s_{ik}\right) \tag{17c}$$

with

$$c_j^{\mathrm{r}} = \max_{\tilde{w}_j\in[0,w_j^{\mathrm{U}}]_{\mathbb{Z}}}\left\{-\sum_{q=1}^{Q}\gamma_{jq}\tilde{w}_j^q\right\} + \max_{\tilde{d}_j\in[d_j^{\mathrm{L}},d_j^{\mathrm{U}}]_{\mathbb{Z}}}\left\{-\sum_{q=1}^{Q}\rho_{jq}\tilde{d}_j^q\right\}, \tag{17d}$$

$$c_{jk}^{\mathrm{t}} = \max_{\tilde{w}_j\in[0,w_j^{\mathrm{U}}]_{\mathbb{Z}}}\left\{-c_k^{\mathrm{x}}\tilde{w}_j - \sum_{q=1}^{Q}\gamma_{jq}\tilde{w}_j^q\right\} + \max_{\tilde{d}_j\in[d_j^{\mathrm{L}},d_j^{\mathrm{U}}]_{\mathbb{Z}}}\left\{c_k^{\mathrm{x}}\tilde{d}_j - \sum_{q=1}^{Q}\rho_{jq}\tilde{d}_j^q\right\}, \tag{17e}$$

$$c_i^{\mathrm{p}} = \max_{\tilde{y}_i\in[0,y_i^{\mathrm{U}}]_{\mathbb{Z}}}\left\{-\sum_{q=1}^{Q}\lambda_{iq}\tilde{y}_i^q\right\}, \tag{17f}$$

$$c_{ik}^{\mathrm{s}} = \max_{\tilde{y}_i\in[0,y_i^{\mathrm{U}}]_{\mathbb{Z}}}\left\{-c_k^{\mathrm{x}}\tilde{y}_i - \sum_{q=1}^{Q}\lambda_{iq}\tilde{y}_i^q\right\}. \tag{17g}$$

Then, the approximate DRNS$^{(Q)}$ model can be solved using a similar separation algorithm as Algorithm 1. We provide a detailed derivation for (17) in Online Appendix Q.

We compare DRNS$^{(Q)}$ with the baseline DRNS in out-of-sample simulations using various orders of moments $Q = 1,2,3,4$ and various nurse pool structures, and we report the results in Table 5. From this table, we observe that DRNS$^{(Q)}$ performs better as Q increases. This is expected because the ambiguity set incorporates more distributional information of $(\tilde{w},\tilde{y})$ as Q increases. In addition, although we obtain an approximate solution to DRNS$^{(Q)}$, the solution eventually outperforms the baseline DRNS as Q increases to three or four. This demonstrates the value of incorporating higher-order moment information, as well as the tightness of the approximation.

Table 5. Comparison of the Out-of-Sample Average Costs Between DRNS and the DRNS$^{(Q)}$ Models

	DRNS	DRNS$^{(1)}$	DRNS$^{(2)}$	DRNS$^{(3)}$	DRNS$^{(4)}$
Structure 0	**33,996**	**33,996**	**33,996**	**33,996**	**33,996**
Structure 1	25,268	32,808	24,956	**24,821**	**24,821**
Structure D	31,051	33,424	32,607	**29,864**	**29,864**
Structure C	25,706	32,833	27,867	24,901	**24,264**
General	24,912	32,822	25,174	**24,574**	**24,574**

Note. The lowest cost for each nurse pool structure is highlighted in bold.

6. Conclusions

We studied a two-stage (DRNS) model for nurse staffing under both exogenous demand uncertainty and decision-dependent absenteeism uncertainty. We derived a min-max reformulation for (DRNS) under general nurse pool structures, leading to a separation algorithm that provably finds a globally optimal solution within a finite number of iterations. Under practical pool structures including one pool, disjoint pools, and chained pools, we derived MILP reformulations for (DRNS) and significantly improved the computational efficacy. Through numerical case studies, we found that modeling absenteeism improves the out-of-sample performance of staffing decisions, and such improvement is positively correlated with the value of flexibility.

For future research, it complements the DRNS model to penalize the nurse shortage on a relative scale, for example, $\sum_{j=1}^{J}(\tilde{d}_j - \tilde{w}_j)^+/\tilde{w}_j$, which penalizes the shortage more heavily when there are fewer nurses showing up. This reflects the increased workload on the remaining nurses when absence occurs.

Endnote

[1] When the NPR requirement is not mandated, we interpret the temporary nurses as a measure of nurse shortage, which may decrease the quality of care and increase the workload of the remaining nurses.

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